PROVE 255: STRICT RECIPROCITY ON LG VIA LAKSHMIBAI–SESHADRI PATHS, WITHOUT BUCH–RAVIKUMAR

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Title	PROVE 255: $\mathrm{SR}_{\mathrm{LG}}$ via LS paths (BR-free), with the Q39 locator result and the registry promotion
WIP commit	`eebd6e3` (`grandpa-rick/work-in-progress`): registry sync, scripts in `scripts/prove255/`. This PDF and its source are added in a follow-up commit.
Proof file	`grandpa-rick/rick-research` commit `1ba5f8f`, `proofs/2026-10-25-crownF-strict-reciprocity.md` §11: `https://github.com/grandpa-rick/rick-research/blob/1ba5f8f23261f4238cb774263d30a5661d4fff29/proofs/2026-10-25-crownF-strict-reciprocity.md`
Registry	`q488-oc-SR_LG-via-LS-paths`: computed → **proved** (commit `c272fed`).

Grades use my ladder: hunch < sketched < computed < checked-sober < proved.

1. What is new

Fix m, a strict partition κ with parts $\le m$, and the Schubert variety $X_\kappa \subset \mathrm{LG}(m, 2m)$ of dimension $|\kappa|$. Let $H_\kappa(j) = \chi(X_\kappa, \mathcal{O}(j))$. Let P_κ be the shifted diagram of κ with the NW product order. Let $F_\kappa(2j)$ be the number of weakly order-preserving maps $G : P_\kappa \to \{0, \dots, 2j\}$ that are even on diagonal boxes. Then

$$H_\kappa(j) = F_\kappa(2j) \quad (j \ge 0),$$

$$(-1)^{|\kappa|} H_\kappa(-j) = N_\kappa(j) := \#\{\text{strict order-preserving } P_\kappa \to [2j-1],\ \text{diagonal values even}\} \quad (j \ge 1).$$

The first identity is proved by a direct bijection with Littelmann's LS paths of shape $j\omega_m$. The second ($\mathrm{SR}_{\mathrm{LG}}$) follows from Ehrhart–Macdonald reciprocity for the order polytope $\mathcal{O}(P_\kappa)$ in the lattice $\Lambda = \mathbb{Z}$ (diagonal coordinates) $\times \frac{1}{2}\mathbb{Z}$ (off-diagonal coordinates). The argument uses no Buch–Ravikumar, no (G0), no hyperplane recursion ($\star$), and not the hand-written §3.3. The proof file therefore contains **two independent proofs of $\mathbf{SR}_{\mathrm{LG}}$**:

- §4: K-theoretic, (G0)+(G1)+(BR);
- §11: LS paths, (G1′)+(LS)+Ehrhart–Macdonald. [proved], given the cited literature.

2. Inputs (corrected citation, Q39)

(G1′) For a Schubert variety $X_w \subset G/P$ and dominant λ with stabiliser P: $H^i(X_w, L_\lambda) = 0$ for $i > 0$, and the geometric Demazure character formula holds: $\mathrm{char}\, H^0(X_w, L_\lambda)^* = \Lambda_w(e^\lambda) := \Lambda_{a_1} \circ \cdots \circ \Lambda_{a_r}(e^\lambda)$ for a reduced word $w = s_{a_1} \cdots s_{a_r}$, where $\Lambda_\alpha(e^\mu) = \frac{e^{\mu+\rho} - e^{s_\alpha(\mu+\rho)}}{1 - e^{-\alpha}} e^{-\rho}$ (Demazure; Joseph; Andersen; Ramanathan; Kumar; Mathieu). This is the same package as (G1).

(LS) *Littelmann's path form of the Demazure character formula is an operator identity:* for every dominant λ and $w \in W/W_\lambda$, $\Lambda_w(e^\lambda) = \sum e^{\pi(1)}$, the sum over LS paths π of shape λ with initial direction $\tau_1 \le w$. *Locator:* P. Littelmann, A Littlewood–Richardson rule for symmetrizable Kac–Moody algebras, Invent. Math. 116 (1994) 329–346, §5.2. It holds for all symmetrizable Kac–Moody algebras, so type C_m is covered.

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Littelmann does *not* prove "char $V_w(\lambda) = \sum e^{\pi(1)}$" directly. That needs (LS) composed with the geometric formula in (G1′). Together they give $h^0(X_w, L_\lambda) = \#\{\text{LS paths of shape } \lambda,\ \tau_1 \le w\}$, which we apply with $\lambda = j\omega_m$. Only this count is used. Weight multisets appear only in the computer check. Earlier versions of §11 cited Ann. of Math. 142 (1995) from memory and stated the character form. Both are corrected. This was a citation-hygiene fix. There was no mathematical gap.

First-hand status of the locator.

- *Read first-hand*, by the owner himself: Littelmann, The path model for representations of symmetrizable Kac–Moody algebras, Proc. ICM Zürich 1994, vol. 1, §8, p. 305, "Demazure type character formula [14]", with [14] = Invent. 116. It was read from the page image (mathunion.org). Verbatim: "For any reduced decomposition $\tau = s_{a_1} \cdots s_{a_r}$ one has $\Lambda_{a_1} \circ \cdots \circ \Lambda_{a_r}(e^\lambda) = \sum_{\eta \in A\pi_\lambda(\tau)} e^{\eta(1)}$". The LS-path definition on p. 304 matches §11's definition up to a harmless sign of β.
- *Secondary:* the section number §5.2 of Invent. 116 is quoted from Chirivì–Fang–Littelmann, arXiv 2207.08904 ("Theorem in Section 5.2 in [L2]"; Littelmann is a coauthor). The Invent. text is paywalled and was not reached.
- Also read (page image): Mathieu, Sém. Bourbaki exp. 798 (1995), Théorème 10, p. 219, which states the same count. The Ann. of Math. 142 (1995) 499–525 paper was not read.

The Bruhat convention ($\tau_1 \le w$ in W/W_λ) and the duality $H^0(X_w, L_\lambda)^* = V_w(\lambda)$ agree with §11. Counts are insensitive to the dual.

3. The proof

Root data. Type C_m, roots $\pm\varepsilon_i \pm \varepsilon_k$, $\pm 2\varepsilon_i$; coroots $(\varepsilon_i \pm \varepsilon_k)^\vee = \varepsilon_i \pm \varepsilon_k$, $(2\varepsilon_i)^\vee = \varepsilon_i$; $\omega_m = (1, \dots, 1)$, $W/W_{\omega_m} \cong \{\pm 1\}^m$. The cell of μ has dimension $d(\mu) = \sum_{\mu_i = -1}(m - i + 1)$, so $\mu \mapsto \kappa(\mu)$ (part set $\{m - i + 1 : \mu_i = -1\}$) is a bijection onto strict partitions with parts $\le m$, with $\dim X_\mu = |\kappa(\mu)|$. For a cover $\kappa_{k-1} = s_\beta \kappa_k$, the *bond* is $|\langle \kappa_k(\lambda), \beta^\vee \rangle|$.

Lemma 11.1 (covers and bonds). *The Bruhat covers in W/W_{ω_m} are exactly "add one box to the shifted diagram". The bond is 1 if the box is on the diagonal (long root $2\varepsilon_m$) and 2 if it is off the diagonal (short root $\varepsilon_i - \varepsilon_k$). Hence the Bruhat order is inclusion of shifted diagrams.*

Proof sketch, full in §11.1. $s_{\varepsilon_i - \varepsilon_k}$ replaces a part q by $p = q + 1$ (a cover iff $p = q + 1$; this is an off-diagonal box with bond $|1 - (-1)| = 2$). $s_{\varepsilon_i + \varepsilon_k}$ raises d by $p + q \ge 3$, so it is never a cover. $s_{2\varepsilon_i}$ adds a part p, which is a cover iff $p = 1$: a diagonal box with bond $|\langle \mu, \varepsilon_m \rangle| = 1$. □

Lemma 11.2. *For $j \ge 1$, $(\tau_1 \supsetneq \cdots \supsetneq \tau_s;\ 0 < a_1 < \cdots < a_{s-1} < 1)$ is an LS path of shape $j\omega_m$ iff for each $i \le s - 1$: $2ja_i \in \mathbb{Z}$, and if $2ja_i$ is odd then $D(\tau_i) \setminus D(\tau_{i+1})$ contains no diagonal box.*

Proof. The bonds of $j\omega_m$ are j times those of ω_m, i.e. jb with $b \in \{1, 2\}$. An a-chain needs $ajb \in \mathbb{Z}$ at each cover, so $2ja \in \mathbb{Z}$. If $ja \in \mathbb{Z}$, any saturated chain works. If $2ja$ is odd, only bond-2 (off-diagonal) covers qualify, and such a chain exists iff the difference has no diagonal box. □

Proposition 11.3 (LS-parity model, BR-free). *LS paths of shape $j\omega_m$ with $\tau_1 \le \kappa$ are in bijection with order-preserving $G : P_\kappa \to \{0, \dots, 2j\}$, even on the diagonal. Hence, given (G1′) and (LS), $H_\kappa(j) = F_\kappa(2j)$ for $j \ge 0$.*

Proof. For $x \in D(\kappa)$ put $g(x) = 2j\, a_{i(x)}$, where $i(x) = \max\{i : x \in D(\tau_i)\}$ ($g = 0$ off $D(\tau_1)$, $a_s = 1$). The $D(\tau_i)$ are down-sets, so g is order-reversing. By Lemma 11.2, g is even on diagonal boxes. Conversely, the distinct interior values $v_1 < \cdots < v_r$ of g give $a_i = v_i/2j$ and $\tau_i = \{g \ge v_i\}$. The two maps are mutually inverse. Put $G = 2j - g$. Count: (G1′) gives $h^0 = \chi$ and $\operatorname{char} H^0(X_\kappa, \mathcal{O}(j))^* = \Lambda_\kappa(e^{j\omega_m})$, and (LS) turns this into the LS-path sum. □

Theorem $\mathbf{SR}_{\mathrm{LG}}$ (second proof). *Given (G1′) and (LS), for all $j \ge 1$: $(-1)^{|\kappa|} H_\kappa(-j) = N_\kappa(j)$, the strict count above.*

Proof. H_κ is a polynomial (Snapper), and $L(j) = \#(j\mathcal{O}(P_\kappa) \cap \Lambda) = F_\kappa(2j)$ is its Ehrhart polynomial. They agree for $j \geq 0$, hence identically. Ehrhart–Macdonald reciprocity gives $L(-j) = (-1)^{|\kappa|}\#(\mathrm{int}(j\mathcal{O}) \cap \Lambda) = (-1)^{|\kappa|}N_\kappa(j)$. □

4. Verification [computed]

- `scripts/prove255/lspaths.py` enumerates LS paths from the root data (covers, bonds and a-chains computed from roots). It checks the full weight multiset against Demazure characters and the count against $F_\kappa(2j)$: $m = 2, 3$ ($j \leq 3$), $m = 4$ ($j \leq 4$), $m = 5$ ($j \leq 2$), 164/164 cases, 0 failures. The bond census gives 0 violations for $m \leq 5$.
- Negative controls that break the bond rule: all bonds 2 fails 21/24 ($m = 3$) and 30/32 ($m = 4$); swapped bonds fails 18/24 and 28/32.
- OG twin (`ogpaths.py`): all type-B bonds are 1, and LS count $= \Omega(P_\kappa, j+1) =$ OG Demazure, 72/72. This gives a BR-free $\mathrm{SR}_{\mathrm{OG}}$ by Stanley reciprocity. The B-analogue of Lemma 11.1 is not written out.

5. Registry promotion

Node `q488-oc-SR_LG-via-LS-paths` moved from computed to **proved** on 2026-10-25 (rick-research `c272fed`). The pre-check passed: its only child `q488-oc-ls-bijection` (Lemmas 11.1, 11.2 and the bijection half of Prop. 11.3) is proved, the node has no `depends_on` edges, and its remaining inputs are cited literature ((G1′), (LS), Ehrhart–Macdonald). The node now records the locator and its first-hand status. Stale "pending (LS)" text in `q488-oc-ls-parity-model` and `q488-oc-even-reciprocity` was updated. The trustcheck validator reports no new problems (61 edges checked; the 48 pre-existing issues are all sources-index format issues).

6. Novelty and honest gaps

(1) **Possible overlap: Dehy 1998 (not read first-hand).** Mathieu, Bourbaki 798, §4, pp. 220–221, reports R. Dehy, *Combinatorial results on Demazure modules*, J. Algebra 205 (1998) 505–524: $X(w) = \{u \leq w\}$ carries a simplicial structure with an integral structure $P_w(n)$, and $P_w(n\lambda) \cong P_w(n)$, so that "les dimensions des modules de Demazure [sont] des polynômes d'Ehrhart". Mathieu adds that it was open whether P_w is always a lattice polytope. This may anticipate the "Demazure dimension = Ehrhart-type count" frame. As far as Mathieu's summary shows, it does not anticipate the LG order-polytope/Λ model or $\mathrm{SR}_{\mathrm{LG}}$. Reading Dehy is now the top novelty item.

(2) Folklore risk for the positive-degree model: De Concini 1979 (symplectic tableaux) is unread.

(3) CCFL 2609.03195 was read first-hand. It is not the owner: it treats chains and lecture hall polytopes only, with no LG Schubert statement and no reciprocity.

(4) The Invent. 116 text itself (§5.2) has not been seen. The statement was read in the owner's ICM survey.

(5) The degeneration remark (Chirivì's semi-toric components ↔ Stanley's canonical triangulation of $(\mathcal{O}(P_\kappa), \Lambda)$) is a [hunch].

7. Why the $\frac{1}{2}$ (Dream 255 addendum)

The two proofs run in opposite directions. §4 goes K-theory → negative side → (Ehrhart–Macdonald) → positive model. §11 goes LS paths → positive model → (E–M) → negative side. So $\mathrm{SR}_{\mathrm{LG}}$ and the positive-degree model are equivalent, and the $\frac{1}{2}$ in the lattice is exactly the bond $\langle \tau\omega_m, \beta^\vee \rangle = 2$ of a short root. This also explains the minuscule/cominuscule split. OG's spinor weight has all bonds 1, so it gets Hibi's order polynomial with no parity. LG's ω_m has short-root bond 2, so it gets the finer lattice. The Crown G dictionary (odd values ↔ primed letters) gets its mechanism here: *a prime is a half-time on a short-root step.* This connects to your reading of $\lceil r/2 \rceil$ as an Ehrhart count of a marked poset. I think it is the same object; see the reply PDF.